15 / 20

20 attempted, 0 left blank · total time on paper 1:14:48

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TMUA SCORE

9.0 2024+ converted 9.0 Official

Official: TMUA 2016's published score conversion. 2024+ converted: our estimate of the equivalent on today's scale.

BY TOPIC

Differentiation	2 / 2
Exponentials and Logar…	0 / 1
General Algebra	0 / 1
General Trigonometry	1 / 2
Geometry	2 / 3
Graphs of Functions	1 / 1
Inequalities	1 / 1
Integration	1 / 2
Logic Counterexample	0 / 2
Logic Deduction	7 / 9
Logic Equivalence	1 / 1
Logic Sufficiency	2 / 2
Sequences and Series	2 / 2

BY DIFFICULTY

3.4	1 / 1
4.3	1 / 1
4.6	1 / 1
4.7	1 / 1
5.4	1 / 1
5.5	1 / 1
5.9	1 / 1
6.0	1 / 2
6.5	1 / 1
6.6	1 / 1
6.7	0 / 2
6.8	2 / 2
6.9	1 / 1
7.0	0 / 1
7.1	2 / 2
8.2	0 / 1

Per-question breakdown

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	A	A	✓	9:57	Bookmark	Review
2	B	B	✓	0:57	Bookmark	Review
3	D	D	✓	1:21	Bookmark	Review
4	C	C	✓	1:43	Bookmark	Review
5	B	C	✗	1:27	Bookmark	Review
6	C	C	✓	2:17	Bookmark	Review
7	H	C	✗	3:09	Bookmark	Review
8	B	B	✓	4:18	Bookmark	Review
9	D	D	✓	2:52	Bookmarked	Review
10	E	E	✓	2:09	Bookmark	Review
11	D	D	✓	3:37	Bookmark	Review
12	F	F	✓	3:01	Bookmarked	Review
13	E	E	✓	1:27	Bookmark	Review
14	B	B	✓	6:28	Bookmark	Review
15	D	D	✓	6:41	Bookmark	Review
16	C	C	✓	4:37	Bookmark	Review
17	F	H	✗	5:37	Bookmarked	Review
18	F	D	✗	6:29	Bookmark	Review
19	F	F	✓	1:31	Bookmark	Review
20	F	E	✗	5:10	Bookmarked	Review

5 B C ✕ 1:27 Bookmark Hide

QUESTION

Consider the statement:

$(*)$ A whole number n is prime if it is 1 less or 5 less than a multiple of 6.

How many counterexamples to $(*)$ are there in the range $0 < n < 50$?

A 2

B 3 YOUR CHOICE

C 4 CORRECT

D 5

E 6

SOLUTION

The correct solution is option C. For details see UAT official worked answers

Logic Counterexample Difficulty 6.7

Missed 1 !!
remember 1 is NOT prime

5 11 17 23 29 35 41 47

1 7 13 19 25 31 37 43

49

QUESTION

The four real numbers a, b, c, and d are all greater than 1.

Suppose that they satisfy the equation $\log_c d = (\log_a b)^2$.

Use some of the lines given to construct a proof that, in this case, it follows that

$(*)\ \log_b d = (\log_a b)(\log_a c)$.

(1) Let $x = \log_a b$ and $y = \log_a c$

(2) $d = (c^x)^2$
(3) $d = c^{(x^2)}$
(4) $d = b^{xy}$

(5) $d = (a^y)^{(x^2)}$
(6) $d = ((a^y)^x)^2$
(7) $d = (a^x)^{xy}$

(8) $d = a^{(y^{2x})}$
(9) $d = a^{(x^2 y)}$

A (1). Then (2), so (6), so (8), so (7), and therefore (4), hence (*) as required.

B (1). Then (2), so (7), so (8), so (6), and therefore (4), hence (*) as required.

C (1). Then (3), so (5), so (9), so (7), and therefore (4), hence (*) as required. CORRECT

D (1). Then (3), so (7), so (9), so (5), and therefore (4), hence (*) as required.

E (1). Then (4), so (5), so (9), so (7), and therefore (3), hence (*) as required.

F (1). Then (4), so (6), so (8), so (7), and therefore (2), hence (*) as required.

G (1). Then (4), so (7), so (8), so (6), and therefore (2), hence (*) as required.

H (1). Then (4), so (7), so (9), so (5), and therefore (3), hence (*) as required. YOUR CHOICE

SOLUTION

The correct solution is option C. For details see UAT official worked answers

Logic Deduction | Exponentials and Logarithms | Difficulty 6.0

QUESTION

Consider these simultaneous equations, where c is a constant:

$$y = 3\sin x + 2$$

$$y = x + c$$

Which of the following statements is/are true?

1 For some value of c: there is exactly one solution with $0 \leq x \leq \pi$ **and** there is at least one solution with $-\pi < x < 0$.

2 For some value of c: there is exactly one solution with $0 \leq x \leq \pi$ **and** there are no solutions with $-\pi < x < 0$.

3 For some value of c: there is exactly one solution with $0 \leq x \leq \pi$ **and** there are no solutions with $x > \pi$.

A	none	**B**	**1** only
C	**2** only	**D**	**3** only
E	**1** and **2** only	**F**	**1** and **3** only — YOUR CHOICE
G	**2** and **3** only	**H**	**1**, **2** and **3** — CORRECT

SOLUTION

The correct solution is option H. For details see UAT official worked answers

Logic Deduction | General Trigonometry | Difficulty 7.0

18 F D ✕ 6:29 Bookmark Hide

QUESTION

Consider this statement about a function $f(x)$:

$(*)$ If $(f(x))^2 \leq 1$ for all $-1 \leq x \leq 1$ then $\int_{-1}^{1}(f(x))^2\,dx \leq \int_{-1}^{1} f(x)\,dx$

Which one of the following functions provides a **counterexample** to $(*)$?

A $f(x) = x + \frac{1}{2}$

B $f(x) = x - \frac{1}{2}$

C $f(x) = x + x^3$

D $f(x) = x - x^3$ CORRECT

E $f(x) = x^2 + x^4$

F $f(x) = x^2 - x^4$ YOUR CHOICE

SOLUTION

The correct solution is option D. For details see UAT official worked answers

Logic Counterexample | Integration | Difficulty 6.7

20 F E ✕ 5:10 Bookmarked Hide

QUESTION

Each interior angle of a regular polygon with n sides is $\frac{3}{4}$ of each interior angle of a second regular polygon with m sides.

How many pairs of positive integers n and m are there for which this statement is true?

A	none	**B**	1
C	2	**D**	3
E	4 CORRECT	**F**	5 YOUR CHOICE
G	6	**H**	infinitely many

SOLUTION

The correct solution is option E. For details see UAT official worked answers

Geometry General Algebra Difficulty 8.2